MILITARY SPECIFICATION

SYM-DI-BETA-NAPHTHYLPARAPHENYLENEDIAMINE

This specification is approved for use within the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture and acceptance of one type of Sym-di-beta-naphthylparaphenylenediamine (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified , the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1218 - ACS Chemicals

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

CODE OF FEDERAL REGULATIONS
Title 49, Transportation, Parts 100-199

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite: 49 CFR 100-199 (latest revision).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E300-73 - Recommended practice for Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

UNITED NATIONS

United Nations - Transport of Dangerous Goods (Recommendations of the Committee of Experts on the Transport of Dangerous Goods)(Application for copies should be addressed to United Nations, Sales Section, First Ave., New York, N.Y.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for assoicated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The sym-di-beta-naphthylparaphenylenediamine shall meet the following requirements when tested in accordance with the applicable test method.

Property	Min.	Max.	Applicable Test Method
Melting Point, ^{o}C	224	230	4.5.1
Weight Loss, (2 hr @ 105°C), %	---	1.0	4.5.2
Fineness (thru No. 100 US Sieve), %	99.5	---	4.5.3
Ash, %	---	0.25	4.5.4

3.2 Color. The Sym-di-beta-naphthylparaphenylenediamine shall be a gray to light tan colored powder, when tested in accordance with 4.5.5.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. The manufacturer shall implement procedures and controls to assure that the process and the product produced are not compromised by foreign materials and contaminants or any other conditions which may degrade the composition. Determination of foreign materials shall be in accordance with 4.5.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for Acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall be ten (10) pounds of Sym-di-beta-naphthylparaphenylenediamine. The sample shall be obtained from the first production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspection to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in Table II (see MIL-A-48078).

4.3.3 Rejection. See MIL-A-48078.

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-S-70692 (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Melting Point			3.1	4.5.1
	Weight Loss			3.1	4.5.2
	Fineness			3.1	4.5.3
	Ash			3.1	4.5.4
	Color			3.2	4.5.5
	Workmanship			3.4	4.5.6

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of one or more batches of Sym-di-beta-naphthylparaphenylenediamine produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of Sym-di-beta-naphthylparaphenylenediamine that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. The lot shall comply with the provisions for submission of product as specified in MIL-STD-105. The criteria and procedure for the assignment of lot numbers shall be in accordance with MIL-STD-1168. Also, MIL-A-48078 applies.

4.4.2 Examination. See MIL-A-48078.

a. Workmanship. Examination for workmanship shall be conducted on each sample selected for testing in accordance with 4.4.3.1. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected.

4.4.3 Testing. PRECAUTION: This specification covers sampling and testing of toxic and hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing the Sym-di-beta-naphthylparaphenylenediamine.

4.4.3.1 Sampling. Approximately 500 grams of the composition shall be selected from each batch to be sampled using ASTM Method E300-73 for solids. Samples shall be selected for inspection in accordance with table below. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. Each required test shall be performed in duplicate. No composite samples shall be used. The classification of defects shall be as given in Table III.

Number of containers in batch	Number of containers to be sampled
2 - 100	2
100 - 200	5
200 - 300	8
300 - 500	10
500 - 1000	13
1000 - 3000	20
3000 - 10,000	32

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TABLE III. Classification of defects.

Test Examination	Paragraph	Defects Classification
Melting Point	3.1	Major
Weight Loss	3.1	Major
Fineness	3.1	Major
Ash	3.1	Major
Color	3.2	Major
Workmanship	3.4	Major

4.4.4 Inspection equipment. For the performance of all tests and examinations specified in 4.4 and 4.5, commercial inspection equipment should be employed. The contractor shall have available and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibrations procedures are followed.

4.5 Methods of inspections. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks. Also, unless otherwise specified herein, all chemicals shall be Reagent Grade of ACS Grade in accordance with MIL-STD-1218. See 6.3 for use of equivalent test methods.

4.5.1 Melting point. The melting point shall be determined as follows:

4.5.1.1 Apparatus.

a. Fisher-Johns melting point apparatus.
b. Optical cover glass.

4.5.1.2 Procedure.

a. Press a small quantity of the sample between two optical cover glasses. Place the glasses (with sample) on the heating block of a Fisher-Johns melting point apparatus.

b. Set the temperature control at "30".

c. Observe the sample through the small magnifying glass provided with the equipment.

d. When the sample is seen to soften (begin to melt) note the temperature and record to the nearest 0.1°C.

e. Continue observing the sample until all solid has turned to liquid. Again note the temperature and record to the nearest 0.1°C.

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f. Report the two temperatures described above as the melting range.

4.5.2 Weight loss. Weight loss shall be determined in accordance with the following:

Weigh approximately 5 to 10 grams of sample accurate to 0.1 mg into a previously dried and weighed weighing bottle. Place in 105 ± 5°C oven for 2 hours. Remove from oven, cool in desiccator and weigh.

$$\% \text{ Weight loss} = \frac{(A - B)\ 100}{A}$$

Where:

A = weight of sample before heating, gm
B = weight of sample after heating, gm

4.5.3 Fineness. Fineness shall be determined in accordance with the following:

Weigh approximately 100 gm of sample accurate to 0.01 gm into a 1000 ml beaker. Add 400 ml deionized water and 100 ml of a 10% Tergitol No. 4 solution. Stir for 3-5 minutes. Pour the resulting suspension onto a No. 100 US Sieve, rinsing the beaker thoroughly with deionized water. Brush gently, breaking soft lumps and aggregates. Rinse with deionized water. Place screen into a 105°C oven for an hour. Cool, weigh residue.

Calculation:

$$\% \text{ through No. 100 Screen} = 100 - \frac{A \times 100}{B}$$

Where: A = weight residue (gm)
B = weight sample (gm)

4.5.4 Ash. Ash shall be determined in accordance with the following:

Weigh approximately 3 gm accurate to 0.1 mg of sample into a previously heated and cooled 30 ml porcelain crucible. Heat gently with a Fisher burner or equivalent until volatiles have been driven off. Transfer crucible to a muffle furnace at 800°C for 1 hour. Cool in desiccator.

Calculation:

$$\% \text{ Ash} = \frac{\text{Weight of Residue} \times 100}{\text{Weight of Sample}}$$

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4.5.5 Color. The sample shall be examined for conformance with proper color.

4.5.6 Workmanship. Visually examine the sample for contamination or foreign matter.

5. PACKAGING.

5.1 Packaging and Packing.

5.1.1 Level A. When applicable packaging shall be as specified in the contract.

5.1.2 Level B. Unless otherwise specified in the contract, packaging and packing shall be in accordance with the standard commercial practice applicable to the type of material. The packaging and packing shall be of such construction and materials that the contents will be adequately protected against damage, loss, or contamination when shipped from the contractor or supply source to the receiving activity, or stored for one year from the date of manufacture. Container size/capacity shall be as specified in the contract. The type of containers and packing materials selected shall comply with applicable carrier rules and regulations. In addition, any hazardous material must be packed and shipped in accordance with the Code of Federal Regulations (CFR), Title 49, Parts 100-199, and also when offered for overseas shipment, the United Nations-Transport of Dangerous Goods (Recommendations of the Committee of Experts on the Transport of Dangerous Goods). It shall be the responsibility of the supplier to obtain all data necessary from DOT, and if applicable, from the Committee of Experts on the Transport of Dangerous Goods (UN-Transport of Dangerous Goods) to ship hazardous materials legally.

5.1.3 Level C. Same as Level B, except not authorized for overseas shipment and that the United Nations-Transport of Dangerous Goods does not apply.

5.2 Marking. In addition to any special marking required by the contract, each container shall be marked in accordance with MIL-STD-129, the Code of Federal Regulations, (CFR) Title 49, Parts 100-199, and the United Nations-Transport of Dangerous Goods (Recommendations of the Committee of Experts on the Transport of Dangerous Goods). Marking shall include but not be limited to, the following information:

a. Title, Number and Date of this Specification

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b. Manufacturer's Name and Location

c. Material Trade Name

d. Net Weight and Volume

e. Lot Number, Batch Number(s), and Date of Manufacture

f. Storage Conditions/Requirements

g. Toxicity Precautions, if applicable

h. Contract Number

i. Warranty Provisions

6. NOTES

6.1 Intended use. The Sym-di-beta-naphthylparaphenylenediamine is intended for use in the M864 propellant.

6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Equivalent test methods. The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, AMCCOM, ATTN: AMSMC-QAR-R, Picatinny Arsenal, NJ 07806-5000. This description should include, but not be limited to, the procedures used, the accuracy and precision of the method; test data to demonstrate the accuracy and precision and drawings of any special equipment required (see MIL-I-45208).

6.4 Subject term (key word) listing.

Propellant
Antioxidant
Projectile

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 6810-A014)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-70692	SYM-DI-BETA-NAPHTHYLPARAPHENYLENEDIAMINE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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